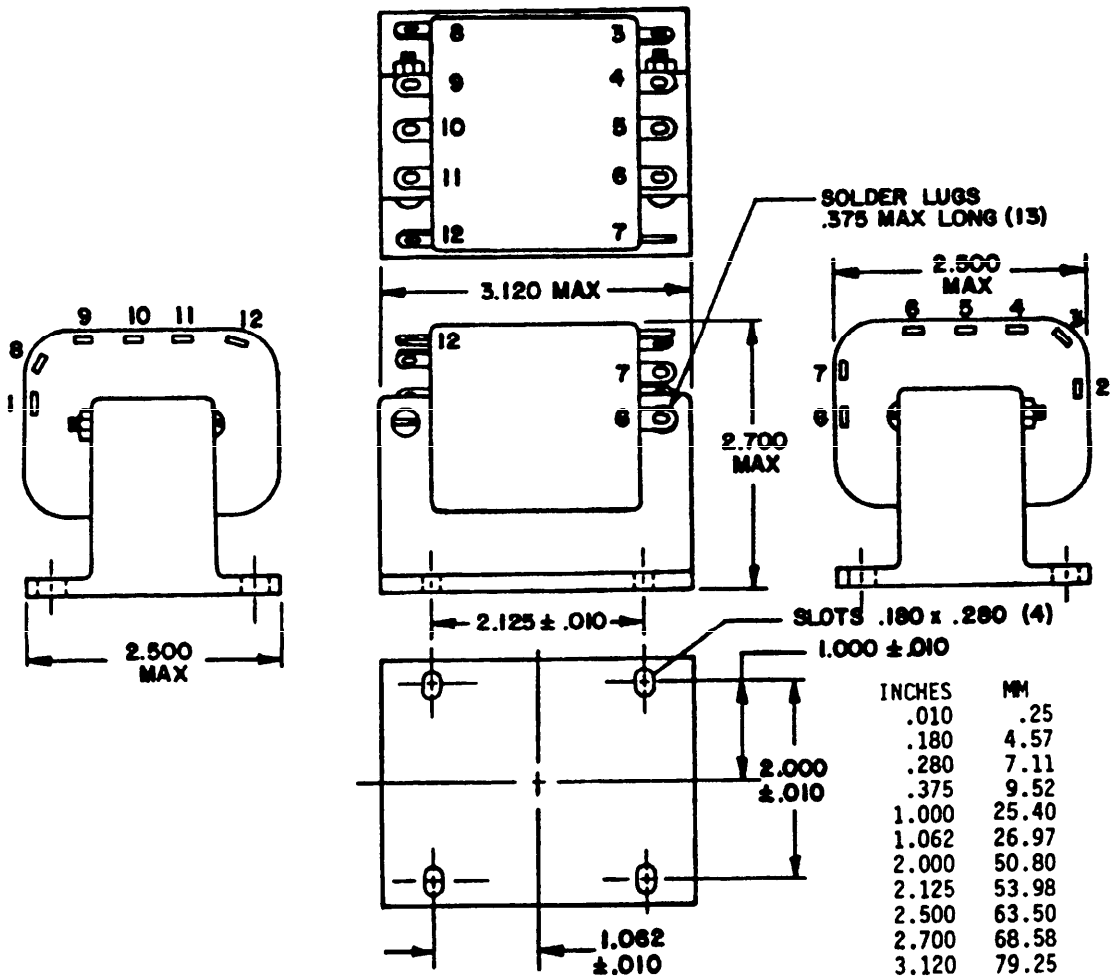
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-UP, TF5S03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.

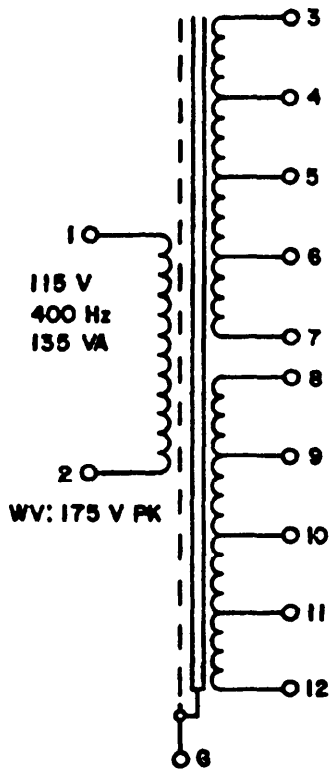


INCHES	MM
.010	.25
.180	4.57
.280	7.11
.375	9.52
1.000	25.40
1.062	26.97
2.000	50.80
2.125	53.98
2.500	63.50
2.700	68.58
3.120	79.25

FIGURE 1. Dimensions and configuration.

FSC 5950

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(3-5-7) 1000 Vrms ct, 40 mAdc
(4-5-6) 650 Vrms ct, 125 mAdc
(8-10-12) 1000 Vrms ct, 40 mAdc
(9-10-11) 700 Vrms ct, 2.25 mAdc
WORKING VOLTAGE (PEAK):
(3-4-5-6-7): 1000 V
(8-9-10-11-12): 1000 V

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on top of transformer.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS: (When numbers are in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
Primary (1-2): 115 volts rms, 400 hertz, 135 volts ampere.
Secondary:
(3-5-7): 1,000 volts rms center tap, ±4 percent, 40 milliamperes dc.
(4-5-6): 650 volts rms center tap, ±4 percent, 125 milliamperes dc.
(8-10-12): 1,000 volts rms center tap ±4 percent, 40 milliamperes dc.
(9-10-11): 700 volts rms center tap ±4 percent, 2.25 milliamperes dc.
Working voltage:
Primary (1-2): 175 volts peak.
Secondary (3-4-5-6-7): 1,000 volts peak.
Secondary (8-9-10-11-12): 1,000 volts peak.

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Fosterite encapsulation.
Terminals: Solder lug terminals, brass, tinned.
Terminal height: .375 inch.
Weight: 2.25 pounds.
Ambient temperature range (operating): -55°C to +101°C.
Operating temperature: 130°C maximum.
Altitude: 10,000 feet operating.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage (at sea level):
Primary (1-2): 500 volts rms.
Secondary (3-4-5-6-7): 2400 volts rms.
Secondary (8-9-10-11-12): 2400 volts rms.

Electrical characteristics:
No load: With 115 volts rms, 400 hertz across (1-2), current in (1-2) 250 milliamperes maximum, power in (1-2) 6 watts maximum.
Voltage across (3-5-7): 1,020 volts rms center tap ±2 percent.
Voltage across (4-5-6): 664 volts rms center tap ±2 percent.
Voltage across (8-10-12): 1,020 volts rms center tap ±2 percent.
Voltage across (9-10-11): 714 volts rms center tap ±2 percent.
Rated load: With 115 volts rms, 400 hertz in (1-2).
Voltage across (3-5-7): 1,000 volts rms center tap ±4 percent at 40 milliamperes dc.
Voltage across (4-5-6): 650 volts rms center tap ±4 percent, at 125 milliamperes dc.
Voltage across (8-10-12): 1,000 volts rms center tap ±4 percent, at 40 milliamperes dc.
Voltage across (9-10-11): 700 volts rms center tap ±4 percent, at 2.25 milliamperes dc.
DC resistance:
(1-2): 1.1 ohms maximum.
(3-4): 48 ohms maximum.
(4-6): 49 ohms maximum.
(6-7): 58 ohms maximum.
(8-12): 280 ohms maximum.

Electrostatic shielding: Voltage ratio shall be 5:1 minimum at 20 kilohertz.

Temperature rise: 29°C maximum with 115 volts rms, 400 hertz, across (1-2) at an ambient temperature of 101°C.

Marking location: See figure 1.

Part number: M27/188-01.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - MU, WC, ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-27/188 TRANSFORMER, POWER, STEP-UP, TF5S032Z

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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